

Full System Database Roadmap - All 12 Tables

1. CUSTOMERS TABLE (Consignors & Consignees)

Stores all senders, receivers, and billing entities. Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
CustomerID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
CustomerType	VARCHAR(20)	NOT NULL	E.g., Consignor, Consignee, Broker.
FullName	VARCHAR(100)	NOT NULL	Registered Name. Mandatory.
GSTIN	VARCHAR(15)	NULL ALLOWED	Leave empty for unregistered clients. Fill for B2B.
ContactPhone	VARCHAR(15)	NOT NULL	Primary contact number.
Address	TEXT	NOT NULL	Complete physical address.

2. VEHICLES TABLE

Registry of all trucks. Connects to LrBilty, EWayBill, Tracking, Fuel, Toll.

Column Name	Data Type	Constraint	Null Rules & Details
VehicleID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
RegistrationNo	VARCHAR(20)	NOT NULL	Number plate (e.g., MH12AB1234). Mandatory.
OwnerName	VARCHAR(100)	NOT NULL	Name of the vehicle owner / agency.
VehicleType	VARCHAR(30)	NOT NULL	E.g., Open Body, Container, Trailer.
LoadCapacity	DECIMAL(10,2)	NOT NULL	Max weight in tons.
RCEpiryDate	DATE	NULL ALLOWED	Leave empty if not tracked. Fill to prevent dispatch.
DriverID (FK)	INT	NULL ALLOWED	Assigned driver. NULL if not assigned.

3. DRIVERS TABLE

Roster of drivers. Connects to LrBilty, Vehicles.

Column Name	Data Type	Constraint	Null Rules & Details
DriverID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
DriverName	VARCHAR(100)	NOT NULL	Full name. Mandatory.
LicenseNumber	VARCHAR(30)	NOT NULL	License code. Mandatory.
LicenseExpiry	DATE	NOT NULL	Expiry date. Mandatory.
Phone	VARCHAR(15)	NOT NULL	Contact number.
Address	TEXT	NULL ALLOWED	Driver address.
Status	VARCHAR(10)	NOT NULL	Active/Inactive. Default Active.

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4. USERS / STAFF TABLE

Access management. Connects to LrBilty (Created by), Tracking (Updated by).

Column Name	Data Type	Constraint	Null Rules & Details
UserID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
Username	VARCHAR(50)	NOT NULL	Login name. Mandatory.
PasswordHash	VARCHAR(255)	NOT NULL	Hashed password. Mandatory.
Role	VARCHAR(20)	NOT NULL	Admin, Manager, Clerk.
IsActive	BOOLEAN	NOT NULL	Default TRUE. FALSE when resigned.

5. FUEL TABLE

Fuel expenses. Connects to Vehicles.

Column Name	Data Type	Constraint	Null Rules & Details
FuelID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
VehicleID (FK)	INT	NULL ALLOWED	Vehicle that was fueled.
Litres	FLOAT	NOT NULL	Amount of fuel.
PricePerLitre	FLOAT	NOT NULL	Cost per litre.
Date	DATE	NOT NULL	Fuel date.
Photo	VARCHAR(100)	NULL ALLOWED	Receipt photo path.

6. TOLL TABLE

Toll expenses. Connects to Vehicles.

Column Name	Data Type	Constraint	Null Rules & Details
TollID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
VehicleID (FK)	INT	NULL ALLOWED	Vehicle that paid toll.
TollName	VARCHAR(100)	NOT NULL	Toll location name.
Amount	FLOAT	NOT NULL	Toll amount.
Date	DATE	NOT NULL	Toll date.
Photo	VARCHAR(100)	NULL ALLOWED	Receipt photo path.

7. INVOICES & PAYMENTS TABLE

Payments ledger. Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
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InvoiceID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
LrNumber (FK)	VARCHAR(30)	NOT NULL	Bilty number. Mandatory.
PaymentType	VARCHAR(20)	NOT NULL	Advance or Balance.
AmountPaid	DECIMAL(10,2)	NOT NULL	Amount received.
PaymentDate	DATETIME	NOT NULL	Transaction time.
TransactionRef	VARCHAR(100)	NULL ALLOWED	UTR for online, empty for cash.

8. TRACKING TABLE

Movement log. Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
TrackingID (PK)	INT	NOT NULL	Auto-increment unique ID. Mandatory.
LrNumber (FK)	VARCHAR(30)	NOT NULL	Bilty number. Mandatory.
CurrentLocation	VARCHAR(200)	NOT NULL	Checkpoint/city.
Status	VARCHAR(50)	NOT NULL	Created, In Transit, Delivered.
UpdateTime	DATETIME	NOT NULL	Timestamp.
Remarks	TEXT	NULL ALLOWED	Optional remarks.
UpdatedBy (FK)	INT	NOT NULL	UserID who updated.

9. LR_BILTY TABLE

Central contract. Connects to Customers, Vehicles, Drivers, EWayBill.

Column Name	Data Type	Constraint	Null Rules & Details
LrNumber (PK)	VARCHAR(30)	NOT NULL	Manual/auto ID. Mandatory.
LrDate	DATE	NOT NULL	Receipt date.
ConsignorID (FK)	INT	NOT NULL	Sender ID.
ConsigneeID (FK)	INT	NOT NULL	Receiver ID.
Origin	VARCHAR(100)	NOT NULL	Start location.
Destination	VARCHAR(100)	NOT NULL	End location.
MaterialDesc	VARCHAR(500)	NOT NULL	Goods details.
TotalWeight	DECIMAL(10,2)	NOT NULL	Weight.
TotalPackages	INT	NOT NULL	Package count.
FreightAmount	DECIMAL(10,2)	NOT NULL	Total fare.
AdvanceAmt	DECIMAL(10,2)	NULL ALLOWED	Pre-paid amount.

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BalanceAmt	DECIMAL(10,2)	NULL ALLOWED	Pending amount.
VehicleID (FK)	INT	NULL ALLOWED	Assigned vehicle.
DriverID (FK)	INT	NULL ALLOWED	Assigned driver.
EWayBillNo (FK)	VARCHAR(20)	NULL ALLOWED	Govt bill number.
DeliveryDate	DATETIME	NULL ALLOWED	Completion date.
Status	VARCHAR(30)	NOT NULL	In Transit, Delivered.

10. EWAYBILL TABLE

Govt link. Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
EWayBillNo (PK)	VARCHAR(20)	NOT NULL	Govt number. Mandatory.
EWayBillDate	DATETIME	NOT NULL	Generation time.
LrNumber (FK)	VARCHAR(30)	NOT NULL	Bilty number.
GenByGSTIN	VARCHAR(15)	NOT NULL	Generator GST.
SupplierGSTIN	VARCHAR(15)	NOT NULL	Sender GST.
RecipientGSTIN	VARCHAR(15)	NOT NULL	Receiver GST.
DispatchFrom	VARCHAR(200)	NOT NULL	Origin address.
ShipTo	VARCHAR(200)	NOT NULL	Destination address.
Distance	INT	NOT NULL	Transit KM.
TotalInvoiceVal	DECIMAL(10,2)	NOT NULL	Cargo value.
CustomDutyVal	DECIMAL(10,2)	NULL ALLOWED	For international.
CGSTAmount	DECIMAL(10,2)	NULL ALLOWED	Tax amount.
SGSTAmount	DECIMAL(10,2)	NULL ALLOWED	Tax amount.
IGSTAmount	DECIMAL(10,2)	NULL ALLOWED	Tax amount.
VehicleNumber	VARCHAR(20)	NOT NULL	Reg number.
TransporterID	VARCHAR(50)	NULL ALLOWED	Outsourced ID.
ValidUpto	DATETIME	NOT NULL	Expiry time.

11. TRIP MANAGEMENT TABLE

Trip records. Connects to Vehicles, Drivers, LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
TripID (PK)	INT	NOT NULL	Auto-increment ID.

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VehicleID (FK)	INT	NOT NULL	Vehicle used.
DriverID (FK)	INT	NOT NULL	Driver assigned.
StartDate	DATETIME	NOT NULL	Trip start.
EndDate	DATETIME	NULL ALLOWED	Trip end.
Origin	VARCHAR(100)	NOT NULL	Start location.
Destination	VARCHAR(100)	NOT NULL	End location.
Status	VARCHAR(20)	NOT NULL	Ongoing, Completed.
LrNumbers	TEXT	NULL ALLOWED	Associated bilty numbers.

12. INVENTORY MANAGEMENT TABLE

Stock tracking. Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
InventoryID (PK)	INT	NOT NULL	Auto-increment ID.
ItemName	VARCHAR(100)	NOT NULL	Item description.
Quantity	INT	NOT NULL	Stock quantity.
Unit	VARCHAR(20)	NOT NULL	Kg, Pieces, etc.
LrNumber (FK)	VARCHAR(30)	NULL ALLOWED	Associated bilty.
Location	VARCHAR(100)	NOT NULL	Storage location.
LastUpdated	DATETIME	NOT NULL	Update timestamp.